HW 8
Stock Options

Problem 11.12.
The price of a non-dividend paying stock is $19 and the price of a three-month European call option on the stock with a strike price of $20 is $1. The risk-free rate is 4% per annum. What is the price of a three-month European put option with a strike price of $20?

Problem 11.19.
The price of a European call that expires in six months and has a strike price of $30 is $2. The underlying stock price is $29, and a dividend of $0.50 is expected in two months and again in five months. Interest rates (all maturities) are 10%. What is the price of a European put option that expires in six months and has a strike price of $30?

Problem 11.20.
Explain the arbitrage opportunities in Problem 11.19 if the European put price is $3.

Problem 11.23.
Prove the result in equation (11.7). (Hint: For the first part of the relationship consider (a) a portfolio consisting of a European call plus an amount of cash equal to K *and (b) a portfolio consisting of an American put option plus one share.)*

Problem 11.24.
Prove the result in equation (11.11). (Hint: For the first part of the relationship consider (a) a portfolio consisting of a European call plus an amount of cash equal to $D+K$ *and (b) a portfolio consisting of an American put option plus one share.)*

Problem 11.26.
Use the software DerivaGem to verify that Figures 11.1 and 11.2 are correct.

Problem 11.31.
Suppose that c_1, c_2, *and* c_3 *are the prices of European call options with strike prices* K_1, K_2, *and* K_3, *respectively, where* $K_3 > K_2 > K_1$ *and* $K_3 - K_2 = K_2 - K_1$. *All options have the same maturity. Show that*

$$c_2 \le 0.5(c_1 + c_3)$$

(Hint: Consider a portfolio that is long one option with strike price K_1, *long one option with strike price* K_3, *and short two options with strike price* K_2.*)*

Problem.
Consider an option on a stock when the stock price is $41, the strike price is $40, the risk-free rate is 6%, the volatility is 35%, and the time to maturity is 1 year. Assume that a dividend of $0.50 is expected after six months.

a. *Use DerivaGem to value the option assuming it is a European call.*
b. *Use DerivaGem to value the option assuming it is a European put.*
c. *Verify that put–call parity holds.*
d. *Explore using DerivaGem what happens to the price of the options as the time to maturity becomes very large and there are no dividends. Explain your results.*